MILITARY SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, 30 THRU 140 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SQ. GRID)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

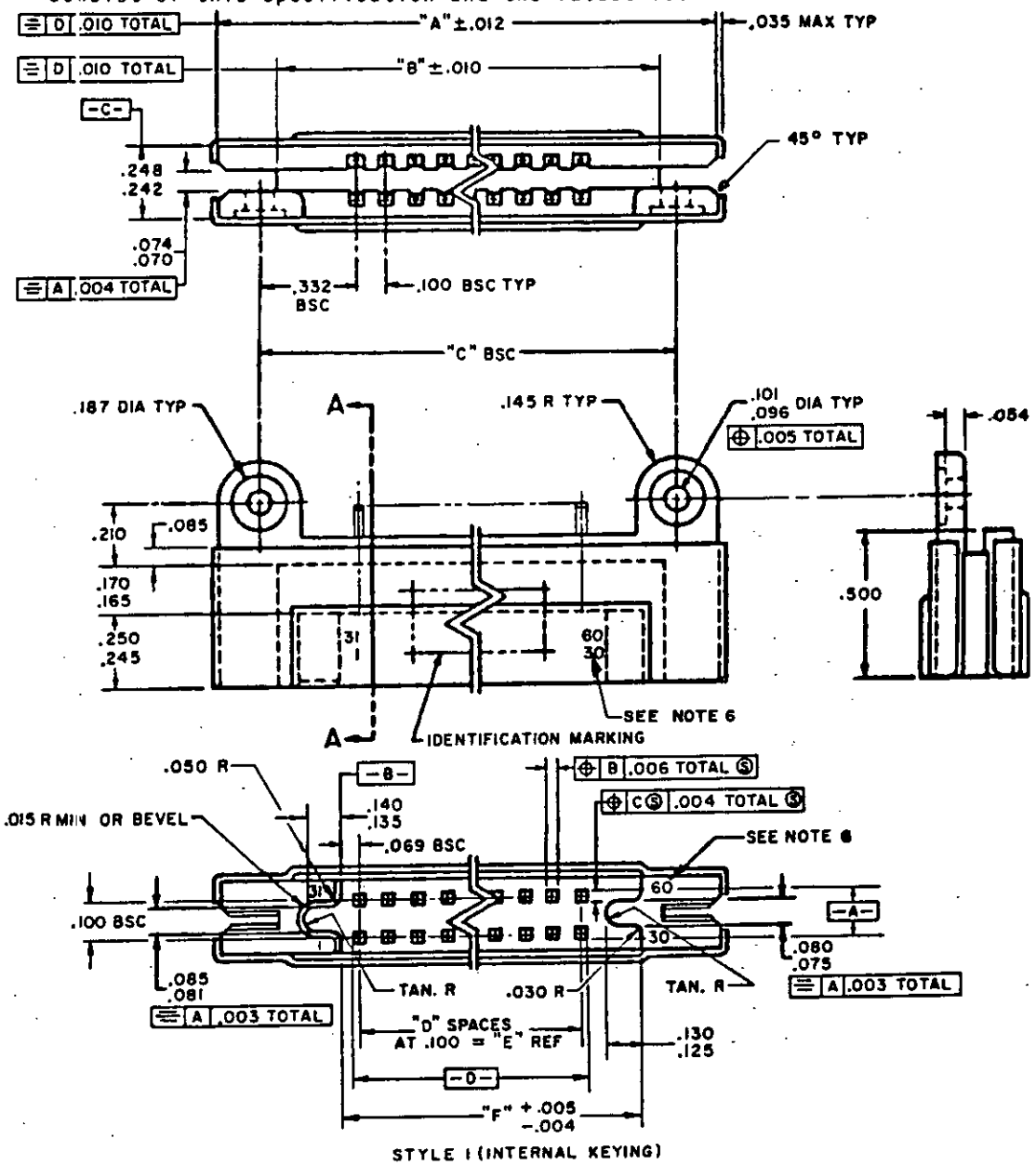



FIGURE 1. Connectors, plug (.100 square grid).
Ⓓ denotes changes


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| Dash number 1/ | Number of contacts | Dimensions | | | | | | Contact identification numbers |
|---|---|---|---|---|---|---|---|---|
| | | A | B | C | D | E | F | |
| 01 | 50 | 3.355 (85.22) | 2.955 (75.06) | 3.065 (77.85) | 24 | 2.400 (60.96) | 2.675 (67.95) | 1 - 25<br>26 - 50 |
| 02 | 60 | 3.855 (97.92) | 3.455 (87.76) | 3.565 (90.55) | 29 | 2.900 (73.66) | 3.175 (80.65) | 1 - 30<br>31 - 60 |
| 03 | 80 | 4.855 (123.32) | 4.455 (113.16) | 4.565 (115.95) | 39 | 3.900 (99.06) | 4.175 (106.05) | 1 - 40<br>41 - 80 |
| 04 | 90 | 5.355 (136.02) | 4.955 (125.86) | 5.065 (128.65) | 44 | 4.400 (111.76) | 4.675 (118.75) | 1 - 45<br>46 - 90 |
| 05 | 100 | 5.855 (148.72) | 5.455 (138.56) | 5.565 (141.35) | 49 | 4.900 (124.46) | 5.175 (131.45) | 1 - 50<br>51 - 100 |
| 06 | 110 | 6.355 (161.42) | 5.955 (151.26) | 6.065 (154.05) | 54 | 5.400 (137.16) | 5.675 (144.15) | 1 - 55<br>56 - 110 |
| 07 | 30 | 2.355 (59.82) | 1.955 (49.66) | 2.065 (52.45) | 14 | 1.400 (35.56) | 1.675 (42.55) | 1 - 15<br>16 - 30 |
| 08 | 40 | 2.855 (72.52) | 2.455 (62.36) | 2.565 (65.15) | 19 | 1.900 (48.26) | 2.175 (55.25) | 1 - 20<br>21 - 40 |
| 09 | 70 | 4.355 (110.62) | 3.955 (100.46) | 4.065 (103.25) | 34 | 3.400 (86.36) | 3.675 (93.35) | 1 - 35<br>36 - 70 |

1/ Dash numbers 01 through 09 are for style 1 (internal keying) connectors.

FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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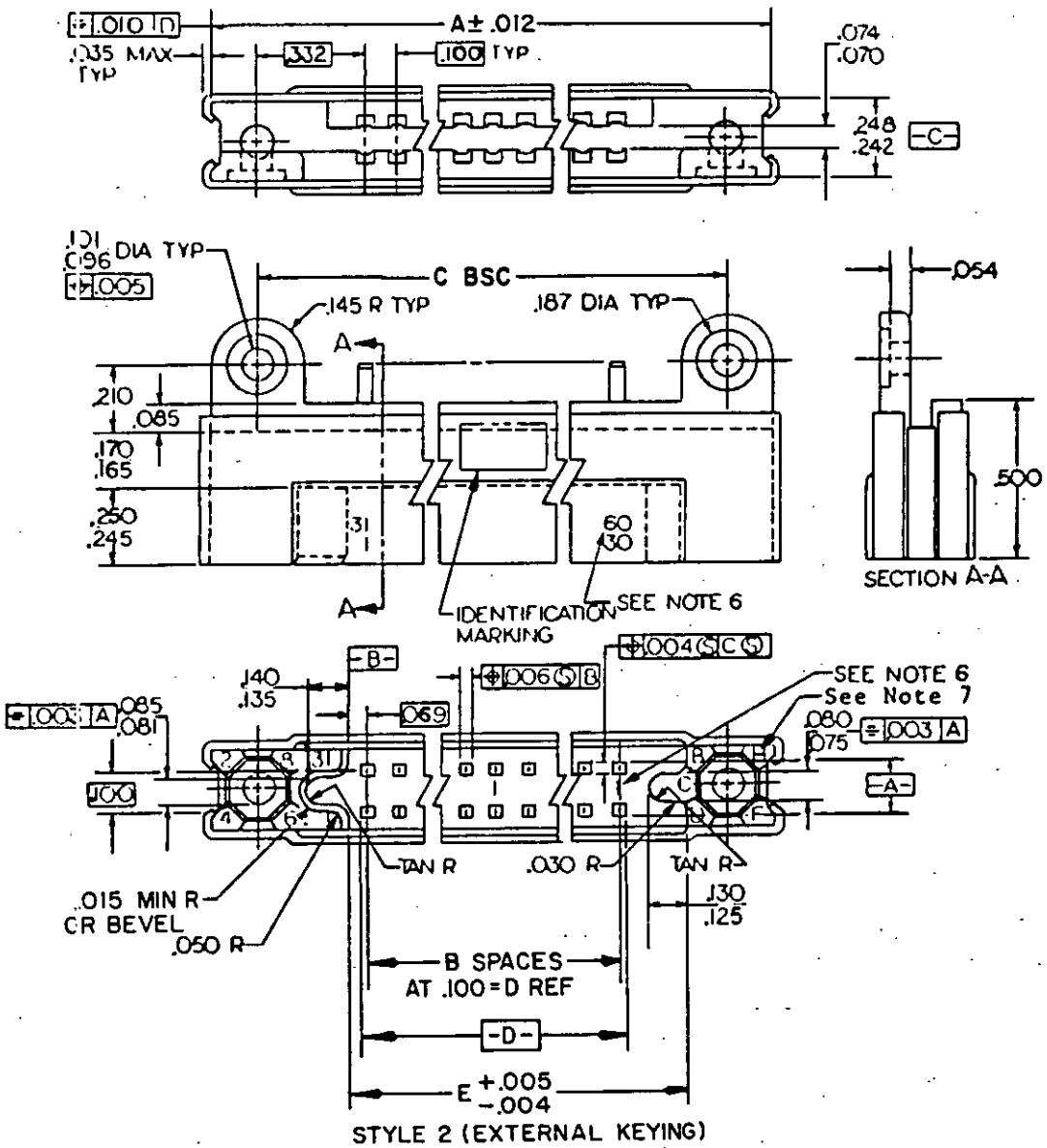



FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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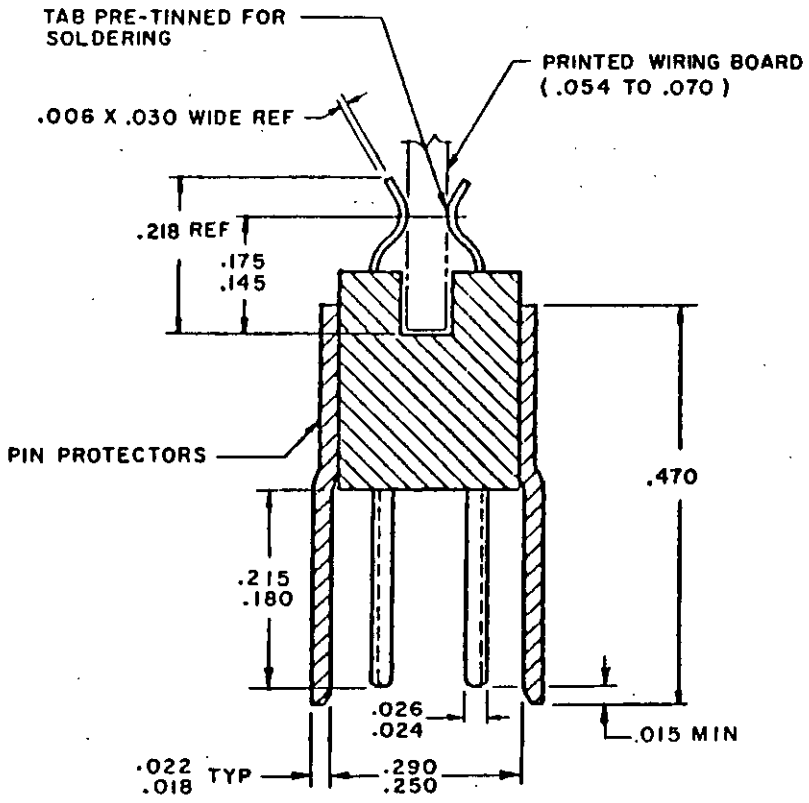
| 1/ Dash number | Number of contacts | Dimensions | | | | | Contact identification numbers |
|---|---|---|---|---|---|---|---|
| | | A | B | D | C | E | |
| 10 | 30 | 2.355 (59.82) | 14 | 1.400 (35.56) | 2.065 (52.45) | 1.675 (42.55) | 1-15 16-30 |
| 11 | 40 | 2.855 (72.52) | 19 | 1.900 (48.26) | 2.565 (65.15) | 2.175 (55.25) | 1-20 21-40 |
| 12 | 50 | 3.355 (85.22) | 24 | 2.400 (60.96) | 3.065 (77.85) | 2.675 (67.95) | 1-25 26-50 |
| 13 | 60 | 3.855 (97.92) | 29 | 2.900 (73.66) | 3.565 (90.55) | 3.175 (80.65) | 1-30 31-60 |
| 14 | 70 | 4.355 (110.62) | 34 | 3.400 (86.36) | 4.065 (103.25) | 3.675 (93.35) | 1-35 36-70 |
| 15 | 80 | 4.855 (123.32) | 39 | 3.900 (99.06) | 4.565 (111.95) | 4.175 (106.05) | 1-40 41-80 |
| 16 | 90 | 5.355 (136.02) | 44 | 4.400 (111.76) | 5.065 (128.65) | 4.675 (118.75) | 1-45 46-90 |
| 17 | 100 | 5.855 (148.72) | 49 | 4.900 (124.46) | 5.565 (141.35) | 5.175 (131.45) | 1-50 51-100 |
| 18 | 110 | 6.355 (161.42) | 54 | 5.400 (137.16) | 6.065 (154.05) | 5.675 (144.15) | 1-55 56-110 |
| 19 | 120 | 6.855 (174.18) | 59 | 5.900 (149.86) | 6.565 (166.75) | 6.175 (156.85) | 1-60 61-120 |
| 20 | 130 | 7.355 (186.82) | 64 | 6.400 (162.56) | 7.065 (179.45) | 6.675 (169.55) | 1-65 66-130 |
| 21 | 134 | 7.555 (191.90) | 66 | 6.600 (167.64) | 7.265 (184.53) | 6.875 (174.63) | 1-67 68-134 |
| 22 | 140 | 7.855 (199.52) | 69 | 6.900 (175.26) | 7.565 (192.15) | 7.175 (182.25) | 1-70 71-140 |

1/ Dash numbers 10 through 22 are for style 2 (external keying) connectors.

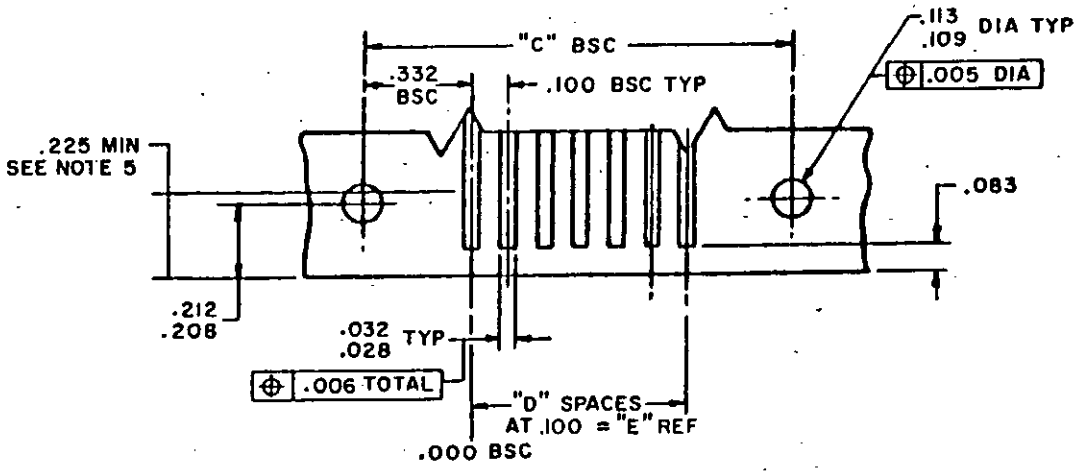
FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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SECTION A-A




RECOMMENDED PRINTED WIRING BOARD LAYOUT

FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | .08 | .101 | 2.57 |
| .004 | .10 | .109 | 2.77 |
| .005 | .13 | .113 | 2.87 |
| .006 | .15 | .125 | 3.18 |
| .008 | .20 | .130 | 3.30 |
| .010 | .25 | .135 | 3.43 |
| .012 | .30 | .140 | 3.56 |
| .015 | .38 | .145 | 3.68 |
| .018 | .46 | .165 | 4.19 |
| .022 | .56 | .170 | 4.32 |
| .024 | .61 | .175 | 4.44 |
| .026 | .66 | .180 | 4.57 |
| .028 | .71 | .187 | 4.75 |
| .030 | .76 | .208 | 5.28 |
| .032 | .81 | .210 | 5.33 |
| .035 | .89 | .212 | 5.38 |
| .050 | 1.27 | .215 | 5.46 |
| .054 | 1.37 | .218 | 5.54 |
| .069 | 1.75 | .225 | 5.72 |
| .070 | 1.78 | .242 | 6.15 |
| .074 | 1.88 | .245 | 6.22 |
| .075 | 1.90 | .248 | 6.30 |
| .080 | 2.03 | .250 | 6.35 |
| .081 | 2.06 | .290 | 7.37 |
| .083 | 2.11 | .332 | 8.43 |
| .085 | 2.16 | .470 | 11.94 |
| .096 | 2.44 | .500 | 12.70 |
| .100 | 2.54 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/27.
5. Minimum pad length on printed circuit board.
6. Numbers indicating, every 5 cavities stamped on side. Numbers indicating end cavities and lines indicating every 5 cavities stamped on mating face.
7. Location indicators embossed on surface.

FIGURE 1. Connectors, plug (.100 square grid) - Continued.



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REQUIREMENTS:

Design and construction:

*Dimensions and configuration:* *See figure 1.*

Material:

Ⓓ Plating: The contact plating for the engagement area, .150 inch minimum length, shall be gold per MIL-C-55302 over nickel per MIL-C-55302. The contact plating for the solder tail area, .160 inch minimum length, shall be tin lead over nickel, with tin lead dip per MIL-C-55302 for .080 minimum length. The remainder of the contact shall be nickel plated per MIL-C-55302.

Contact identification: See figure 1.

Ⓓ Coating: The aluminum pin protector shall be coated per MIL-A-8625, type II or epoxy coated .001 to .003 inch thick.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at 75°F.

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

*Contact resistance:* The average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:
Sea level: 900 volts rms.
High altitude: 200 volts rms.

Part number: M55302/26- (dash number from figure 1).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of *manufacturers of the item either* for the Government or for use in equipment to be delivered to the Government.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - CR, AR, MI
Navy - SH, AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3645-04)



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